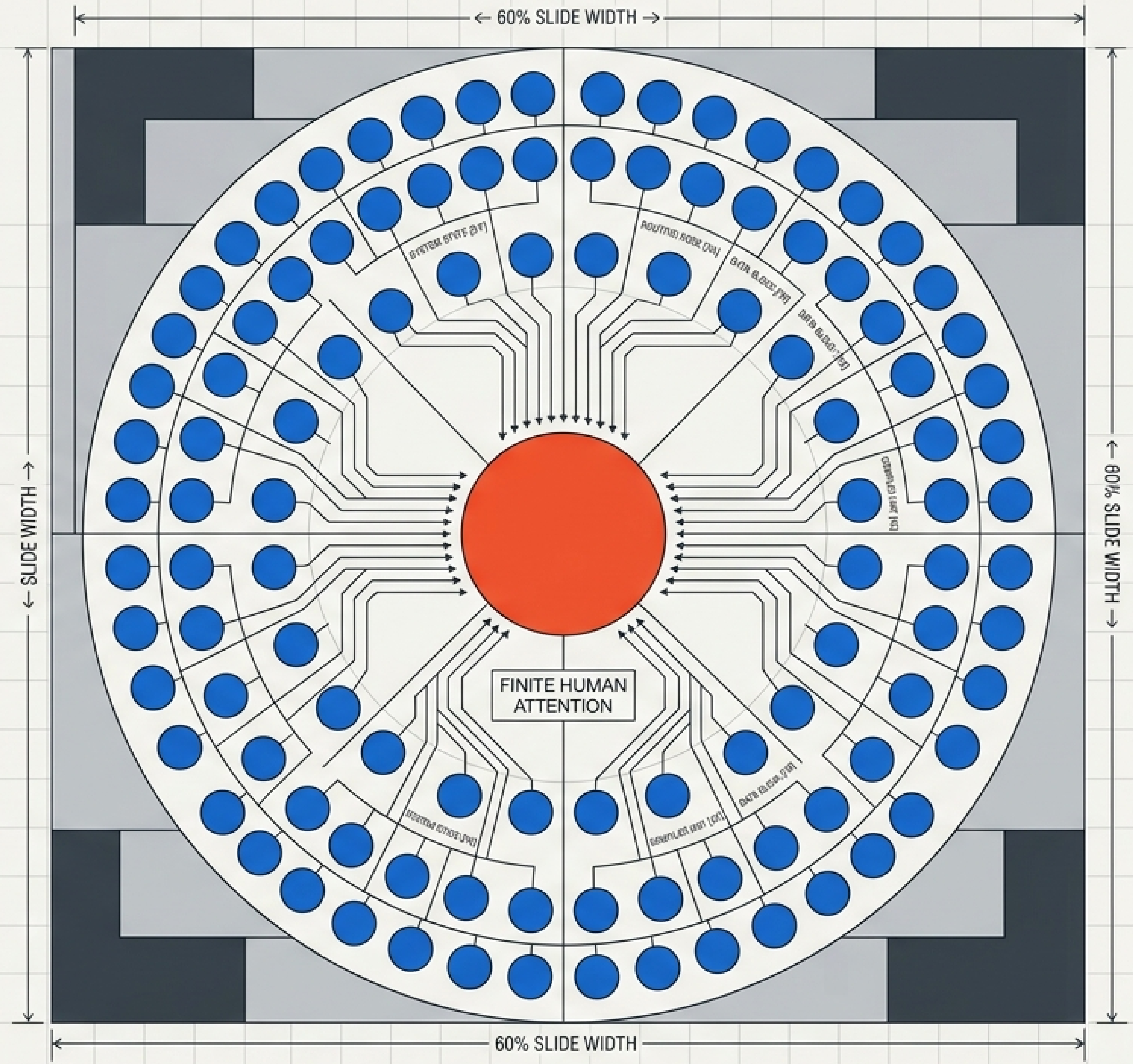
The Attention Compiler

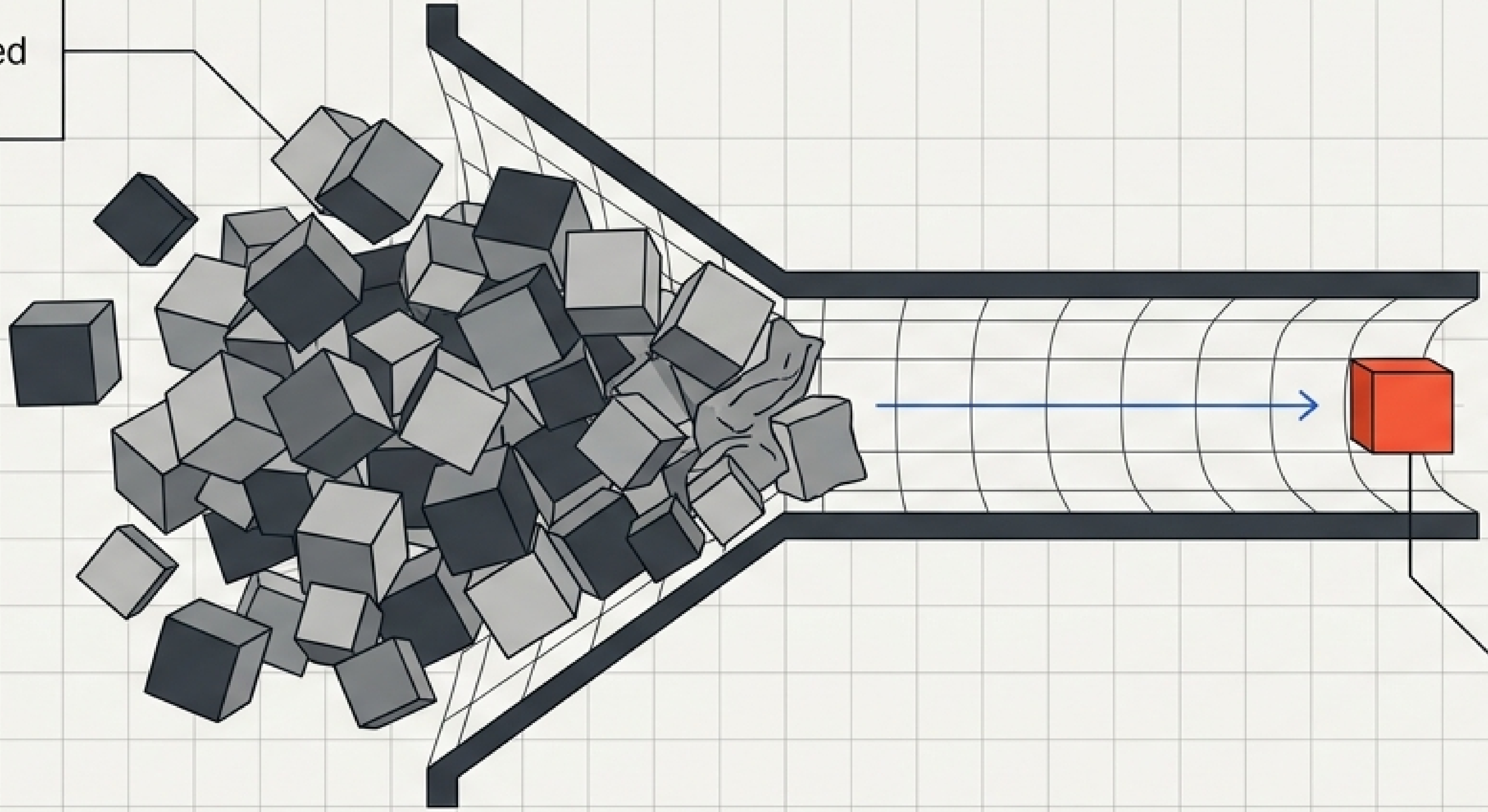
Prerequisite Routing, Adaptive Curricula, and the Triage of Novel Ideas



Based on the 2026 framework by Flyxion
Independent Researcher

The Many-to-One Asymmetry of Conversational Concurrency

The Infinite Supply:
Digital scale allows unlimited inbound communication.



$$\lambda E[C] < A$$

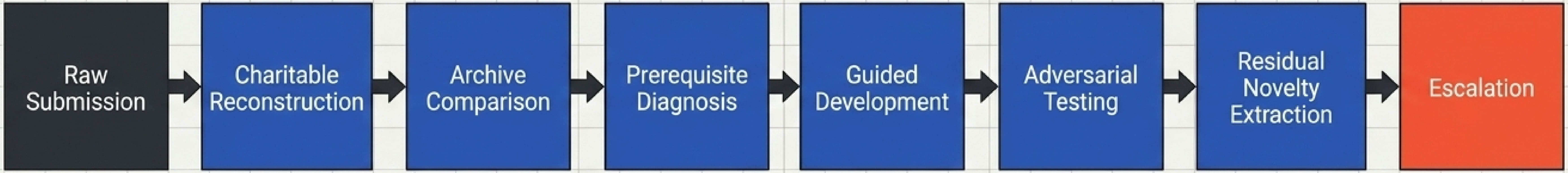
The Fixed Ceiling:
Human attention remains a finite computational resource.

The central failure of the modern inbox is the repeated loss of cognitive labor.
A thinker is forced to restart every conversation from zero, reconstructing the exact same conceptual path for every new arrival.

Moving from Chronological Queues to Epistemic Routing

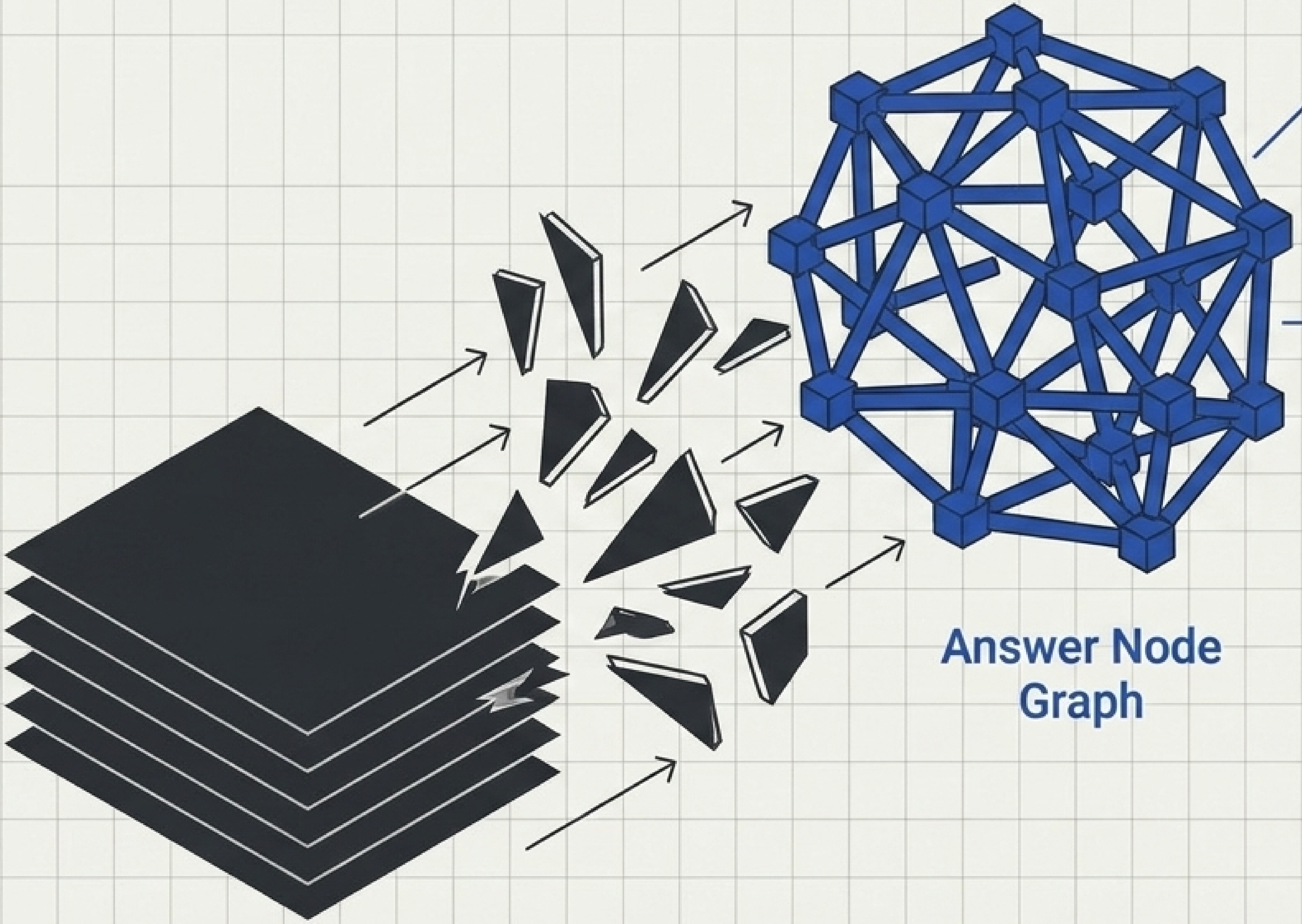
	Traditional Gateway	Attention Compiler
Treatment of Input	Chronological and identical for all.	Prerequisite-sensitive and adaptive.
Valuation Metric	Rewards linguistic polish, institutional credentials, and recognizable formats.	Rewards residual novelty, conceptual transfer, and structural difference.
Function	A passive filter that accepts or rejects.	An active compiler that transforms, reconstructs, and develops.
Systemic Outcome	Information overload or exclusionary walls.	Continuous knowledge synthesis and protected attention.

The Compilation Pipeline



A compiler does not merely filter input. It transforms expressions from one representational level into another—detecting failures and preserving relevant structure—to determine what operations must be applied before an idea's true value can be assessed.

The Archive as Executable History



Computed Memory: Decomposing past work into answer nodes, distinctions, and unresolved problems.

Direct Routing: Recurring questions are instantly routed to the exact point they were previously resolved.

The Provenance Imperative

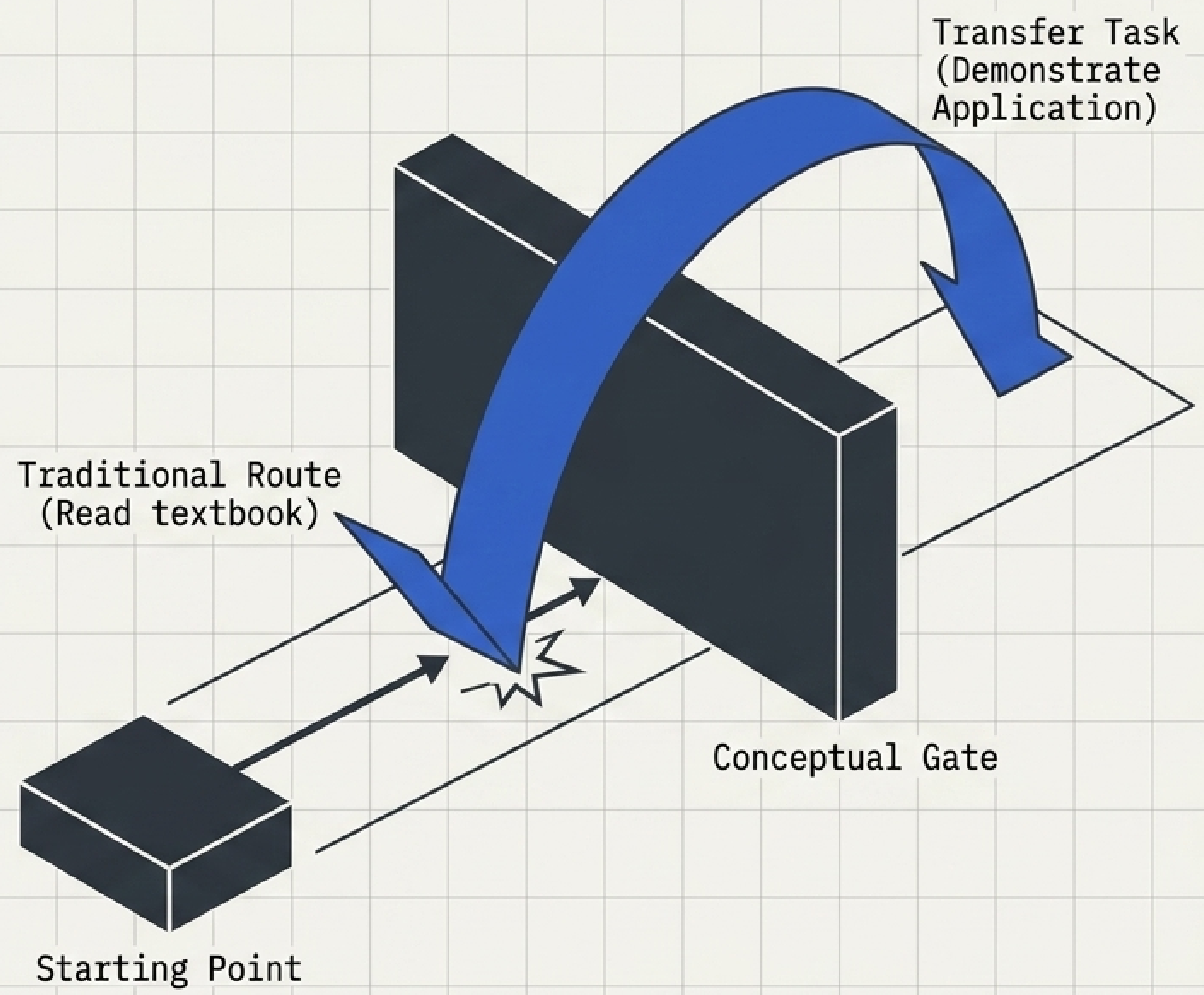
To prevent automated mediation from fabricating positions (AI hallucinations), the system **strictly separates** what the thinker explicitly wrote, what the system inferred, and what the interlocutor contributed.

The Prerequisite Gauntlet

Dynamic Curricula: The system detects missing dependencies (e.g., conflating semantic vs. statistical information) and generates a learning path attached specifically to the submitted idea.

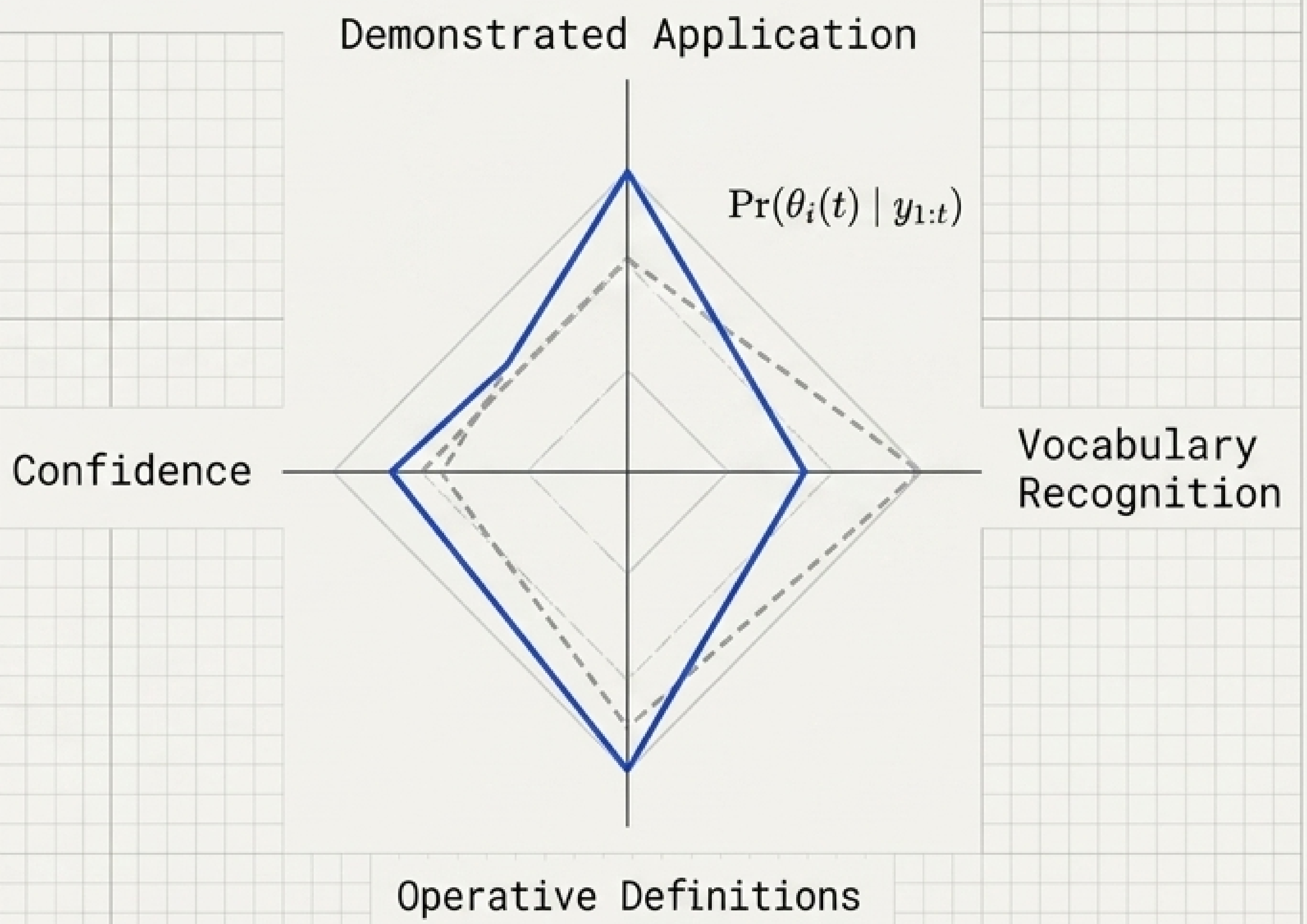
Bypass via Application: Users bypass prerequisites by demonstrating understanding through proof, counterexample, or explanation—not by holding credentials.

Comprehension ≠ Assent: A participant passes a gate by accurately understanding a distinction, not by agreeing with it. Collapsing this difference creates ideological screening.



Estimating Epistemic State

The system maintains a localized, Bayesian model of what the interlocutor knows, calibrated through subtle conversational feedback rather than declarative tests.

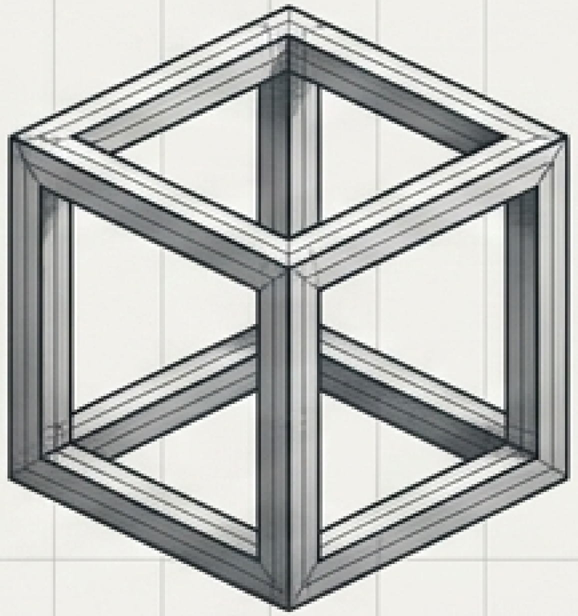


Key Principle: Transfer over Recognition

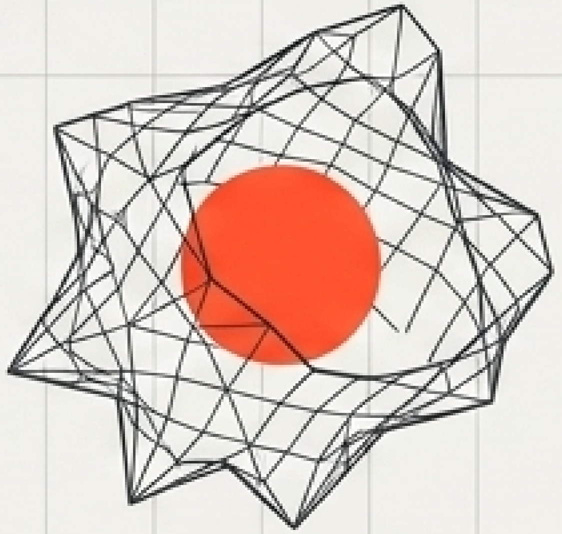
Asking a user if they understand a concept provides zero reliable data. Asking them to **apply a limiting case to their own idea** simultaneously advances the conversation and reveals their true operational capability.

Developmental Triage for Unusual Ideas

Traditional review systems reward proposals that already resemble recognized disciplinary forms—masking unoriginality and punishing genuine, terminologically immature novelty.



Zero Novelty / Disciplinary Polish

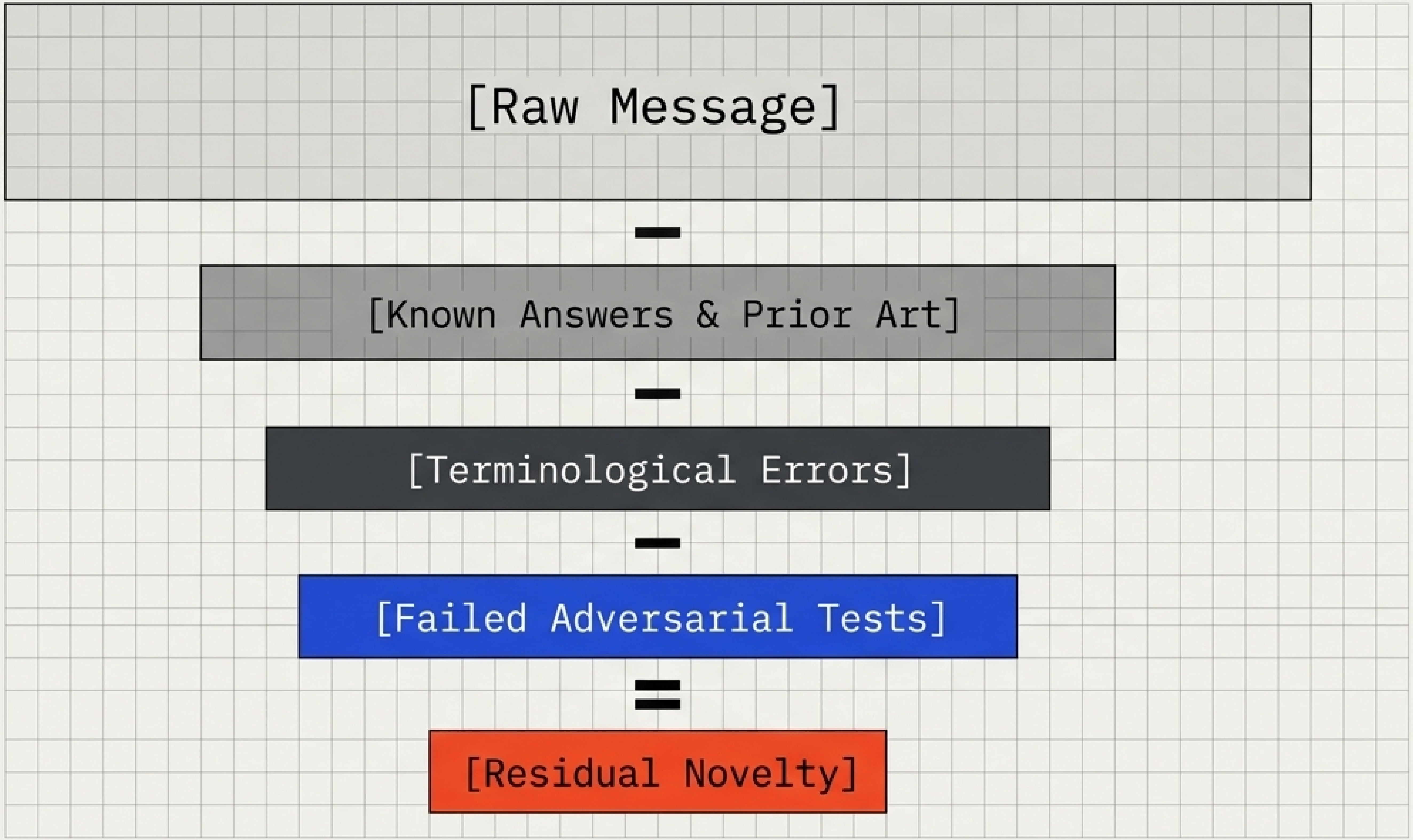


High Novelty / Terminologically Immature

The Triage Workspace:

The compiler provides an adversarial workspace to reconstruct strange ideas. It asks: What does this exclude? What derivations expose its failure? It separates idiosyncratic phrasing from the underlying structural contribution.

Extracting Residual Novelty



Novelty is evaluated after transformation, not before.

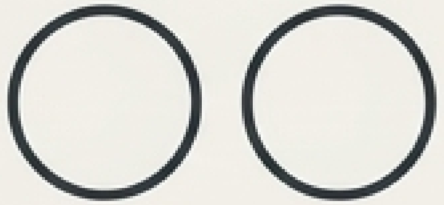
The question is not whether a submission looks unfamiliar on its surface, but whether it falls outside what the existing domain model accounts for.

The Escalation Packet:

The human recipient never sees the raw noise. They receive only the absolute frontier of the conversation: the unresolved remainder, along with the exact history of tests the idea survived to get there.

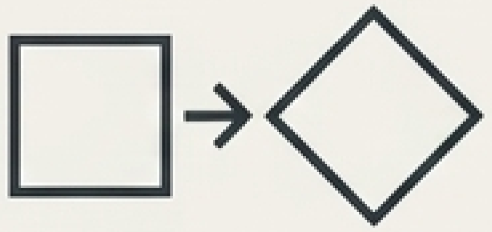
The Multidimensional Taxonomy of Novelty

$$\rho(x) = (\rho_s, \rho_f, \rho_e, \rho_c, \rho_a)$$



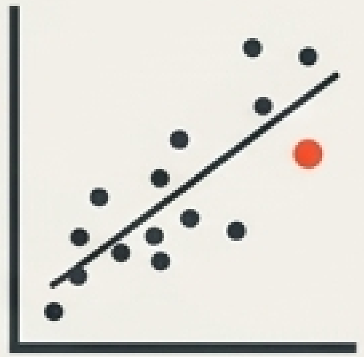
Semantic

Radically new meaning or conceptual content.



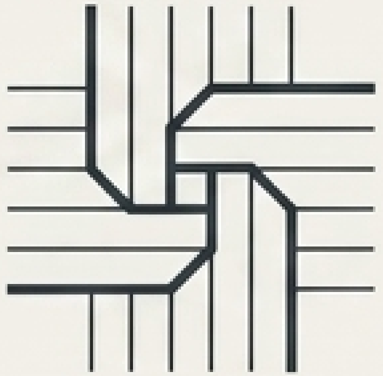
Formal

Equivalent meaning, but structurally new mathematical/logical architecture.



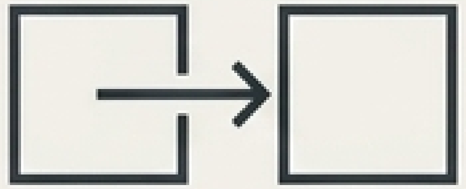
Empirical

New observational data challenging existing models.



Combinatorial

A new, stable synthesis of previously isolated known parts.



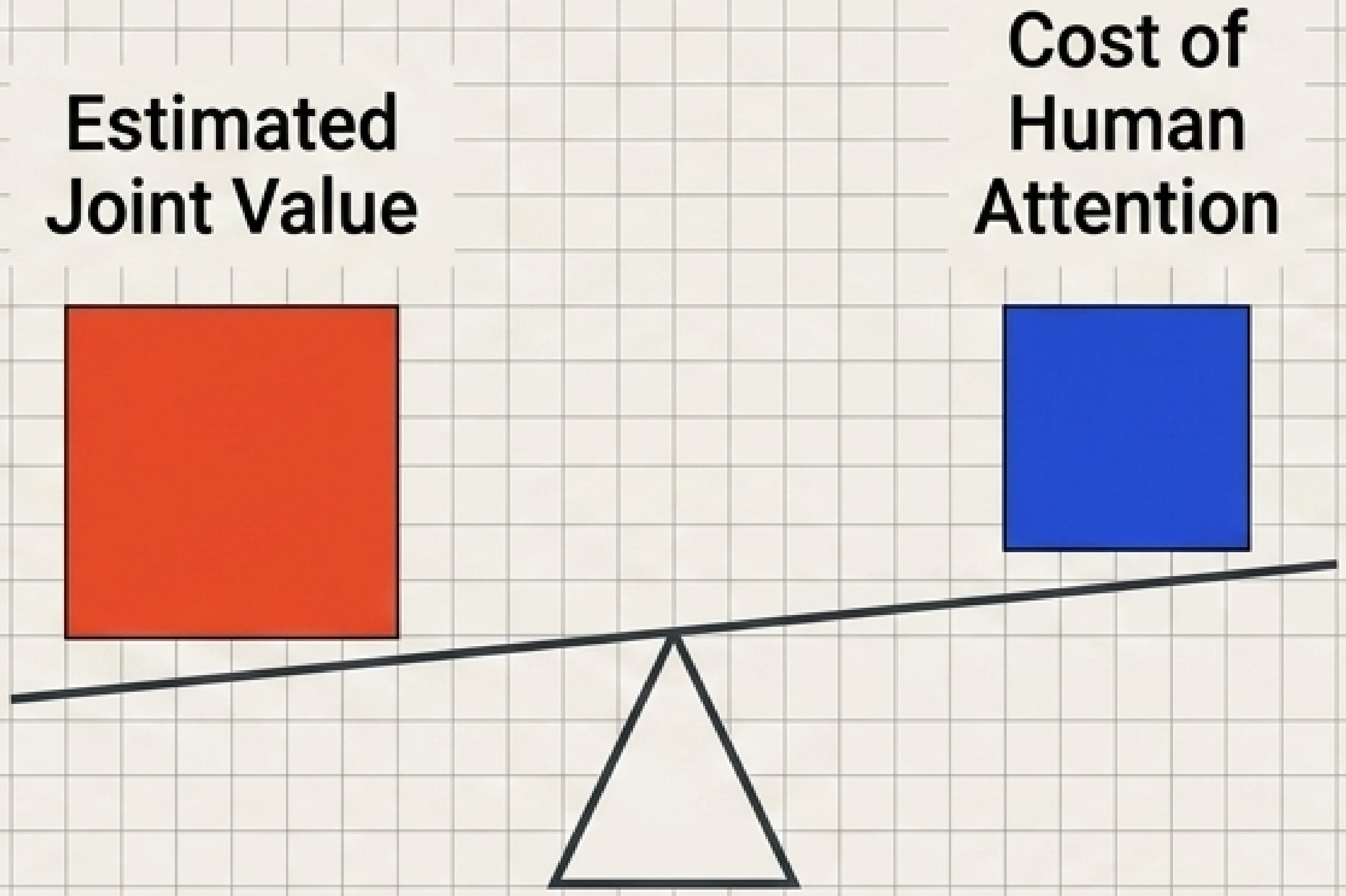
Applicative

A well-known concept successfully transferred to an entirely new domain.

Takeaway: An idea may be mathematically familiar but novel in application, or verbally unrecognizable yet formally identical to an established construct. The compiler detects both.

The Mathematics of Escalation

Escalation is an allocation of scarce resources, not a judgment of personal worth. Importance is general; salience is recipient-relative.



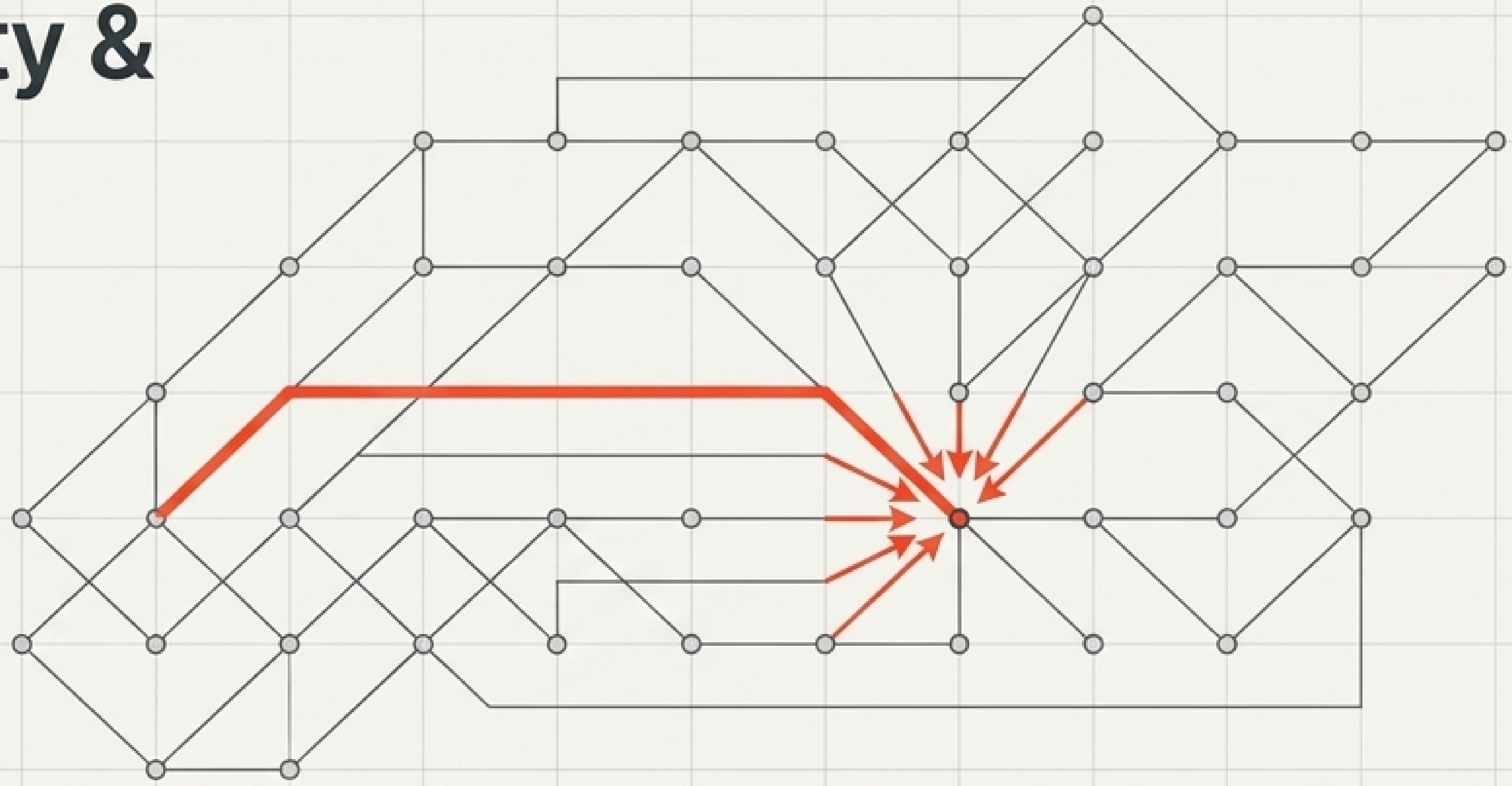
$$\mathbb{E}[\Delta_{\mathrm{joint}}(x_i, r) \mid K_r] > c_i$$

The Threshold:

The system only escalates a submission when the expected intellectual gain from direct, reciprocal human-to-human interaction outweighs the attention cost.

If the system or archive can resolve it lione it alone, it must do so.

Distributional Novelty & The Topology of Misunderstanding



- Semantic novelty evaluates the content of an individual message.
- **Distributional novelty** evaluates the aggregate behavior of thousands of messages.

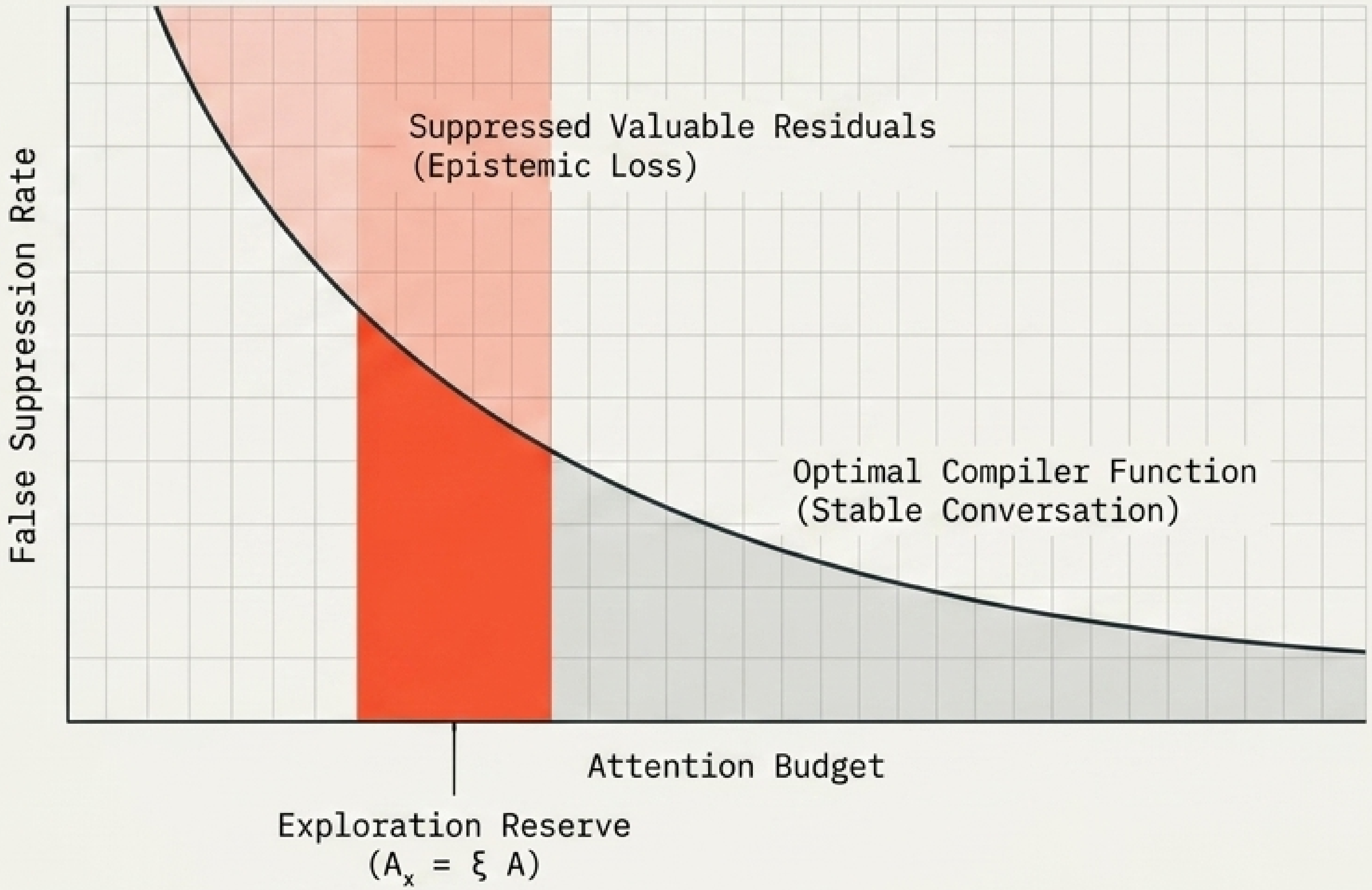
The Insight: If 500 independent participants fail at the exact same conceptual transition, the fault lies not in the users, users, but in the curriculum or the original exposition. Even non-escalated conversations provide vital feedback, mapping the precise topology of public misunderstanding.

Bounding Epistemic Loss

The Danger of Efficiency: A purely greedy algorithm optimizing for easily evaluated ideas will discard difficult, paradigm-shifting proposals.

The Exploration Budget: The system reserves a fixed fraction of attention (A_x) specifically for uncertain classifications and unfamiliar domains.

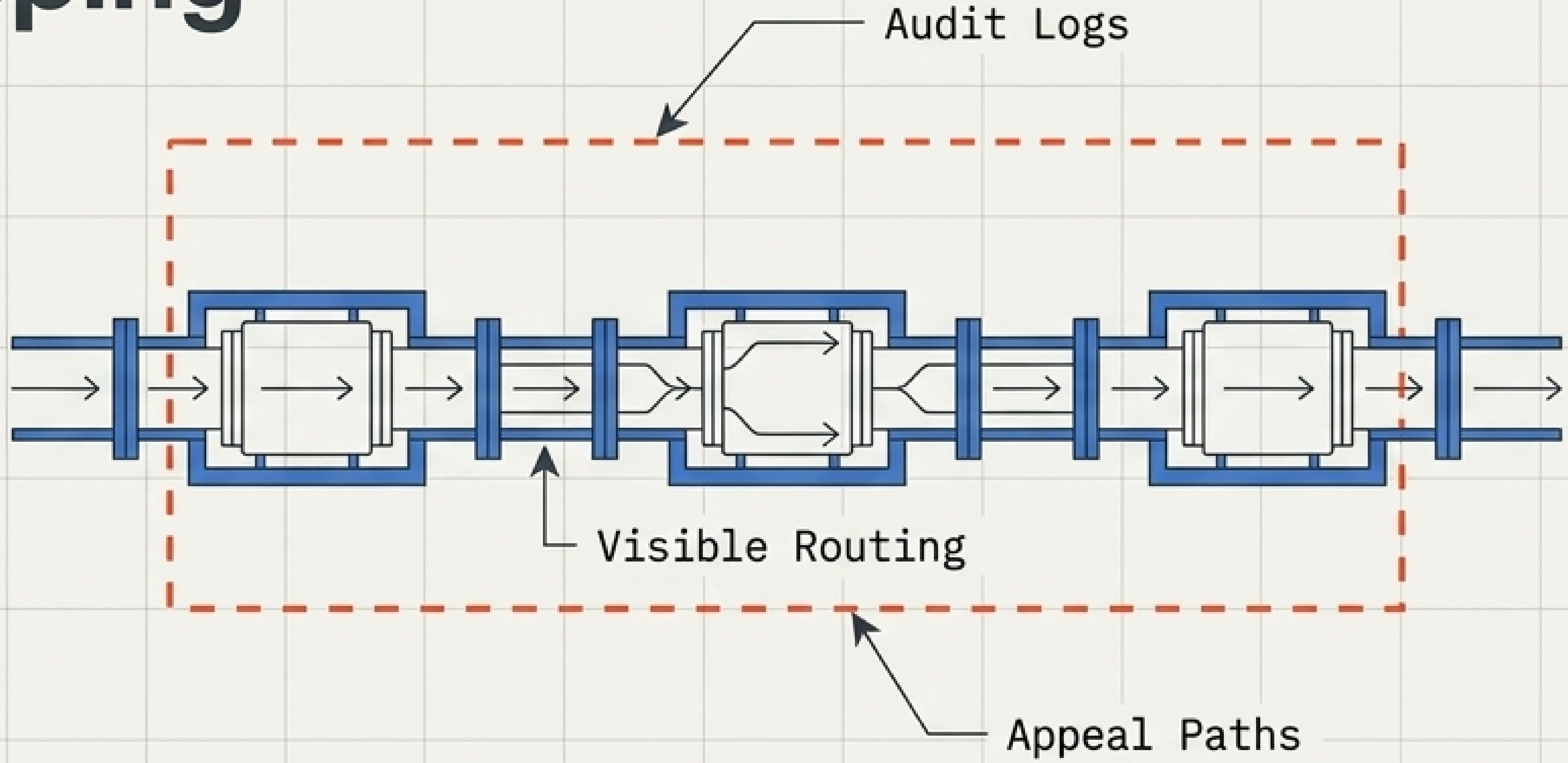
The Optimization Goal: The compiler does not optimize for maximum rejection. It optimizes for stable conversation under finite capacity while aggressively guarding against the false suppression of valuable residuals.



Governance & The Threat of Epistemic Gatekeeping

The Broadcaster's Power: The exclusive right to determine what gets transmitted to many.

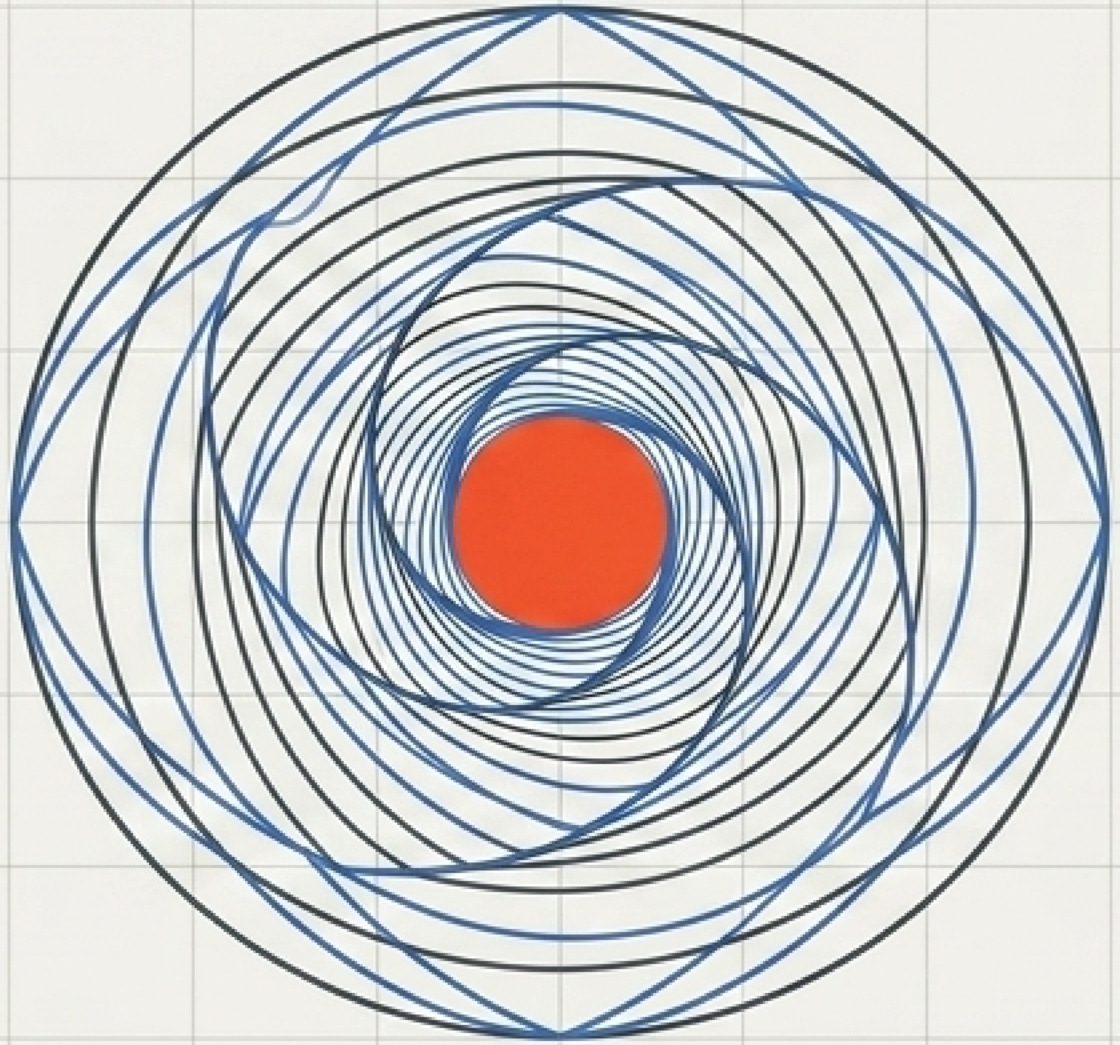
The Filterer's Power: The exclusive right to decide what counts as "adequately said" by the many.



The Safeguard

A system protecting attention can easily become coercive, encoding contested assumptions as neutral prerequisites. The user must maintain the architectural right to declare: "I understand this prerequisite, and I explicitly reject it."

Access Through Transformation



Universal access and finite attention are not mutually exclusive.
The system succeeds when every interaction results in a more precise, answerable, and fulfilled participant—whether they reach the center or not.

Every person may enter, every idea may begin in its native form, and every message must become more intelligible as it approaches the frontier of another person's attention.